1 Full-Mixing Assumption

We first consider a simplest model. We assume propagules produced by a successful strain-2 patch are evenly dispersed across all n patches. Every target patch receives the same average propagule input, and the establishment probability is identical across patches. And we temporarily assume strain 2 always goes extinct if strain 1 exists in the patch (this is more strict than real-world situation), i.e., strain 2 can only establish in the fraction q of empty patches, where q is determined by the strain-1 equilibrium.

2 Establishment Probability

$$\lambda = \kappa z D \left[(1-g)\, N_i^{(k-1,3)} \mathbb{1}_i^{(k-1,3)} + \frac{g}{n} \sum_{j=1}^{n} N_j^{(k-1,3)} \mathbb{1}_j^{(k-1,3)} \right]$$

Let $g = 1$ (to indicate full-mixing) and assume the spread of strain 2 starts from a single patch, we have

$$\lambda_2 = \frac{\kappa_2 z_2 D N^*}{n}$$

The establishment probability is

$$P_2(\text{establish}) = 1 - e^{-\lambda_2 (1-P_{f,2})}$$

2.1 Invasion number

Only the empty fraction q of patches can be colonized by strain 2. We can define patch-level invasion number

$$R_{\text{inv}} = q\, n(1 - e^{-\lambda_2}(1 - P_{f,2}))$$

It is the expected number of newly infected patches by strain 2 caused by the initial infected patch. $R_{\text{inv}}^{\text{mix}} > 1$ indicates successful invasion.

3 Mean-Field Equilibrium of Strain 1

The equilibrium of strain 1 determines q and N^*.

3.1 Equilibrium fraction of infected patches

$$P_1(\text{establish}) = 1 - e^{-\lambda_1 (1-P_{f,1})}$$

where

$$\lambda_1 = \kappa_1 \rho_1 z_1 D N^*$$

Under the same mean-field mixing assumption, the equilibrium patch fraction ρ_1 satisfies

$$\rho_1 = 1 - e^{-(1-P_{f,1})\kappa_1 \rho_1 z_1 D N^*}.$$

Hence

$$q = 1 - \rho_1.$$

3.2 Equilibrium host population size

The mean-field annual update is

$$N_{k+1} = (1 - z_1 D\rho_1)\Big[N_k + rN_k(1 - N_k/K)\Big].$$

Setting $N_{k+1} = N_k = N^*$ yields

$$N^* = K\left[1 - \frac{1}{r}\left(\frac{1}{1 - z_1 D\rho_1} - 1\right)\right].$$

4 Invasion condition

The invasion condition is

$$R_{\text{inv}} > 1,$$

where

$$R_{\text{inv}} = (1 - \rho_1)\, n \left(1 - e^{-\lambda_{\text{mix}}(1 - P_{f,2})}\right), \quad \lambda_{\text{mix}} = \frac{\kappa_2 z_2 D N^*}{n},$$

and (ρ_1, N^*) are determined by the belowing equations

$$\rho_1 = 1 - \exp[-(1 - P_{f,1})\, \beta_1 p_1\, \rho_1 \nu_1 z_1 D N^*]\,,$$

$$N^* = K\left[1 - \frac{1}{r}\left(\frac{1}{1 - z_1 D\rho_1} - 1\right)\right].$$